SAYAM SETHI

Email | Google Scholar | GitHub | LinkedIn

EDUCATION

PhD in Electrical and Computer Engineering — August 2023 – 2028 (tentative)
Advisor: Dr. Jonathan Baker; The University of Texas at Austin — *Austin, Texas*
GPA: **3.90/4.00**

Bachelor of Technology | **Dept. of Computer Science and Engineering** — July 2019 – May 2023
Indian Institute of Technology, Delhi — *Delhi, India*
GPA: **9.55/10.00**

PUBLICATIONS[1]

Staying Sober: A Shot-Free Predictor of Logical Error Rates for Concatenated Quantum Error Correcting Codes — Oct 2026
S. Sethi, M. Poster, A. Awasthi, J. Viszlai, and J.M. Baker — *arXiv*

Exploring Asymmetric QEC Code Concatenation — June 2026
S. Sethi, M. Poster*, A. Awasthi, W. Yang, J. Viszlai, and J.M. Baker* — *QEC '26 (poster)*

Optimizing Parallel Execution of Commuting Pauli Product Rotations — May 2026
S. Sethi, D. Nambisan, and J.M. Baker — *arXiv*

Price and Payoff: Non-Determinism in Fault Tolerant Quantum Computation — May 2026
A. Awasthi, S. Sethi, S. Khan, G.S. Ravi, and J.M. Baker — *QCE '26*

INJEQT: Improved Magic-State Injection Protocol for Fault-Tolerant Quantum Extractor Architectures — April 2026
S. Sethi, S. Khan, A. Awasthi, A. Anand, and J.M. Baker — *QCE '26*

Architecting Early Fault Tolerant Neutral Atoms Systems with Quantum Advantage — April 2026
S. Khan, S. Sethi, K. Sahay, Y. Lin, J. Alnas, S. Kurapati, A. Anand, J.M. Baker, and K.R. Brown — *Under review*

Logical Compilation for Multi-Qubit Iceberg Patches — April 2026
C. Mazzetti, S. Sethi, R. Rines, P. Gokhale, and J.M. Baker — *Under review*

Optimizing Logical Mappings for Quantum Low-Density Parity Check Codes — March 2026
S. Sethi, S. Khan*, M. Poster, A. Anand, and J.M. Baker* — *Under review*

RESCQ: Realtime Scheduling for Continuous Angle Quantum Error Correction Architectures — March 2025
S. Sethi, and J.M. Baker — *ASPLOS '25*

Cyclic Qubit Mappings — September 2024
M. Poster, S. Sethi, and J.M. Baker — *QCE '24*

3D-TemPo: Optimizing 3-D DRAM Performance Under Temperature and Power Constraints — February 2024
S. Pandey, S. Sethi, and P.R. Panda — *IEEE TCAD*

[1] * denotes equal contribution

Internships

Infleqtion May 2026 - August 2026
Graduate Quantum Software Intern

IBM Research, TJ Watson Research Center Summer 2024 and Summer 2025
Quantum Research Scientist Intern

Quadeye Securities LLP, Gurugram Jun – Jul 2022
Systems Intern

Invited Talks and Posters

Exploring Asymmetric QEC Code Concatenation (poster) June 2026
8th International Quantum Error Correction conference *Santa Barbara, CA*

INJEQT: Improved Magic-State Injection Protocol for Fault-Tolerant Quantum Extractor Architectures May 2026
IBM Circuit and Compilers Group *Virtual*

RESCQ: Realtime Scheduling for Continuous Angle Quantum Error Correction Architectures April, May 2025
IBM Circuit and Compilers Group *Yorktown Heights, NY*
UT Austin CompArch Seminar *Austin, TX*

Professional Services

- Artifact Evaluation PC for 59th IEEE/ACM International Symposium on Microarchitecture (MICRO 2026).
- Junior PC for Symposium on Parallelism in Algorithms and Architectures (SPAA 2025).

Academic Achievements

- **Texas Quantum Institute Graduate Fellowship (2025-27)**: Awarded a fellowship for investigating classical overheads in FTQC compilation
- **IIT Delhi Semester Merit Award**: Secured for being in the **top 7 percentile** in **Semester I** of **2019-20** and **2021-22**
- **ICPC India Regionals, Amritapuri '21**: Secured **India Rank 47** and **Institute Rank 1** out of **4393 teams**
- **Google HashCode '22**: Secured **India Rank 12**, **Institute Rank 1** and **Global Rank 240** out of **10177 teams**
- **Indian National Mathematical Olympiad**: Qualified for **INMO 2018** (precursor to IMO) on being shortlisted in the Telangana region in **RMO 2017**
- **Kishore Vaigyanik Protsahan Yojana (KVPY)**: Offered the fellowship in **SX stream (2018-19)** and **SA stream (2017-18)** with **All India Ranks** of **34** and **271** respectively

Teaching Experience and Extra-Curricular Activities

Teaching Assistant

- Grad TA for ECE 412 - Software Design & Implementation in Fall '26, course for ECE sophomores
- Grad TA for ECE 382V - Quantum Error Correction Theory & Systems in Spring '26
- Grad TA for ECE 312 - Software Design & Implementation in Fall '24, Spring '25, course for ECE sophomores
- Grad TA for ECE 306 - Introduction to Computing in Fall '23, introductory course for ECE freshers
- Lead TA for COL216 **(received the outstanding TA award)** - Computer Architecture in Spring '23, mandatory course for CSE sophomores
- Undergraduate TA for COL215 - Digital Logic and System Design in Fall '22, mandatory course for CSE sophomores

Algorithms and Coding Club, IIT Delhi

- **Overall Coordinator, 2022-23**: Elevated the club's recognition from a Chapter to a Technical Club; organized events in Tryst, Technical Fest at IITD
- **Coordinator, 2021-22**: Conducted the Summer of CP and took live lectures on YouTube; guided executives
- **Executive, 2020-21**: Authored various problems for Competitive Programming contests conducted by the club

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Java, VHDL, Assembly, Standard ML, Octave, MATLAB, R
Libraries/Frameworks and Tools: Qiskit, Stim, Sniper, HotSpot, CoMeT, Vivado, LaTeX, git